Terraform Codebase Structure

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REPOSITORY LAYOUT

```
infrastructure-repo/
├─ modules/                          17 modules
│  ├─ network/
│  ├─ security/
│  ├─ database/
│  └─ … 14 more
│     main · variables · outputs · versions
├─ envs/
│  ├─ dev/                           deployed
│  ├─ preprod/                       planned
│  └─ prod/                          planned
├─ docs/                             architecture · diagrams
├─ bitbucket-pipelines.yml
└─ .terraform.lock.hcl
```

envs/dev ROOT MODULE · 27 MODULE CALLS

Foundation	`network` `security`
Data	`database` `db-ops-task` `s3-uploads`
Edge & TLS	`acm-certificate ×4` `alb` `cloudfront-alb ×2` `static-site` `site-form`
Containers	`ecr ×4` `ecs-cluster` `ecs-service ×4`
CI/CD & ops	`ci-oidc` `alarms` `env-scheduler` `budget`

Each layer consumes outputs (IDs, ARNs, endpoints) of the layers above — no hard-coded identifiers.

STATE & GUARDRAILS

- **Remote state**
  Encrypted S3 state per env + native locking
- **Account guard**
  allowed_account_ids: wrong account fails plan
- **Provider alias**
  us-east-1 only for CloudFront certificates
- **Default tags**
  Project · Env · ManagedBy on every resource
- **Input validation**
  account ID, environment, OIDC identifiers
- **Secrets**
  write-only + ephemeral: never in state
- **Versions**
  Terraform ≥ 1.11 · AWS ~> 6.0 · lock file

Same modules, per-environment terraform.tfvars — instance sizes, AZ / NAT count, Multi-AZ, schedules

dev · IMPLEMENTED | preprod · PLANNED | prod · PLANNED